ELEE 5318/6352
SPECIAL PROBLEM –ARRAYS

SP - 2. Develop a code for plotting the magnitude of the array factor in the form of a polar radiation pattern, and calculate the half power beamwidth and sidelobe level for an N-element array with *arbitrary* magnitude and phase distribution. Go back to first principles and find the magnitude of the array factor by summing the original series. Allow each element to have its own magnitude and phase (which is equivalent to a complex excitation coefficient). Make it as general as possible. You will not have a uniform phase shift β in your formulation. The equation to plot is given below and is a modification of Eqn. 6-6 which was for a uniform array. For this assignment you will be able to insert arbitrary values for both the magnitude A_n and the phase ϕ_n for each element in the array. You can simulate a uniform linear array by choosing each $A_n = 1$ for all n and each ϕ_n to have a constant phase shift between adjacent elements.

Do not use the formulas from the book to calculate the beamwidth and sidelobe level. They are only for uniform arrays. Instead use your calculated values for the array factor to iteratively find them. Show examples to illustrate the ability of your program and compare with results from the formulas in the text.

$$AF = \sum_{n=1}^{N} [A_n \, e^{j\varphi_n}] \;\; e^{+j(n-1)[kd\, cos\theta]}$$